

Jeonghoon Lim

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🔗 ADS 📄 Google Scholar 🆔 0000-0003-2719-6640



Employment History

2026-present 📌 **Nevada Center for Astrophysics Postdoctoral Fellow**
University of Nevada, Las Vegas, NV

Education

2020-2026 📌 **Iowa State University, Ames, IA**
Ph.D. in astrophysics: *Dust-gas dynamics and planetesimal formation driven by the streaming instability*; Advisor: Prof. Jacob B. Simon
Graduated: June 2026

2018-2020 📌 **Chungnam National University, Daejeon, South Korea**
M.S. in astronomy: *Statistics of Turbulence Driven by Solenoidal and Compressive Drivings*; Advisor: Prof. Jungyeon Cho

2012-2018 📌 **Chungnam National University, Daejeon, South Korea**
B.S. in astronomy

Research Interests

📌 Planet formation theory, physics of protoplanetary disks, (magneto)hydrodynamic turbulence, computational astrophysics

Awards and Achievements

2022-2025 📌 Future Investigators in NASA Earth and Space Science and Technology (FINESST)

2022-2023 📌 Preparing Future Faculty Associate: Completed a two-semester professional development program offering advanced training in teaching, mentoring, and leadership to prepare graduate students for faculty careers

2020-2022 📌 The Korean Government Scholarship Program for Study Overseas

2020 📌 Graduate Research Award by Journal of the Korean Astronomical Society

Computational Skills/Experience

📌 Godunov Schemes; Particle-mesh; Grid codes

📌 High-performance computing on supercomputers in ACCESS and NASA-HEC

📌 Co-PI on 2022 – 2023 ACCESS Allocation (AST120062, 29 Million CPU Hours, PI: J. Simon, "Numerical Simulations of Planetesimal Formation")

📌 Codes/Programming Languages: Athena, Julia, C, Python

Professional and Committee Services

- 2025  External Reviewer for NASA's Solar System Workings Program
- July 2018  LOC Member, 8th East-Asia School and Workshop on Laboratory Space and Astrophysical Plasmas

Student Mentoring

- Fall 2022 - Spring 2024  Olivia Brouillette, former undergrad at ISU; now in graduate program at New Mexico State University

Teaching Experience

Iowa State University, IA

- Spring 2024  PHYS 132: General Physics II; Astro 150: Star, Galaxies, and Cosmology
- Fall 2023  PHYS 115: Physics for the Life Sciences
- Fall 2020  PHYS 111 Lab: General Physics

Chungnam National University, Daejeon, South Korea

- Spring 2019  300-level Galactic Astronomy

References

Prof. Jacob B. Simon

Associate Professor
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Prof. Wladimir Lyra

Associate Professor
New Mexico State University
wlyra@nmsu.edu
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Prof. Steven Kawaler

Distinguished Professor
Iowa State University
sdk@iastate.edu
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List of Publications and Products

As of Aug. 2026, **6** first-author articles • **10** total publications • **145** total citations on ADS (**125** from first-author papers) • **h-index: 6**

First-author articles

- 1 **J. Lim**, J. B. Simon, R. Li, O. Brouillette, D. G. Rea, and W. Lyra, “The Streaming Instability in 3D: Conditions for Strong Clumping,” *ApJ*, vol. 1000, no. 2, p. 156, Apr. 2026, Art. no. 156.
 DOI: 10.3847/1538-4357/ae47e9
- 2 **J. Lim**, S. A. Baronett, J. B. Simon, C.-C. Yang, D. Sengupta, O. M. Umurhan, and W. Lyra, “Bridging Unstratified and Stratified Simulations of the Streaming Instability for $\tau_s = 0.1$ Grains,” *ApJ*, vol. 993, no. 1, p. 12, Nov. 2025, Art. no. 12.  DOI: 10.3847/1538-4357/ae01a6
- 3 **J. Lim**, J. B. Simon, R. Li, D. Carrera, S. A. Baronett, A. N. Youdin, W. Lyra, and C.-C. Yang, “Probing Conditions for Strong Clumping by the Streaming Instability: Small Dust Grains and Low Dust-to-gas Density Ratio,” *ApJ*, vol. 981, no. 2, p. 160, Mar. 2025, Art. no. 160.  DOI: 10.3847/1538-4357/adb311
- 4 **J. Lim**, J. B. Simon, R. Li, P. J. Armitage, D. Carrera, W. Lyra, D. G. Rea, C.-C. Yang, and A. N. Youdin, “Streaming Instability and Turbulence: Conditions for Planetary Formation,” *ApJ*, vol. 969, no. 2, p. 130, Jul. 2024, Art. no. 130.  DOI: 10.3847/1538-4357/ad47a2
- 5 **J. Lim** and J. Cho, “Decay of Turbulence in Fluids with Polytropic Equations of State,” *Journal of Korean Astronomical Society*, vol. 53, pp. 49–57, Apr. 2020.  DOI: 10.5303/JKAS.2020.53.2.49
- 6 **J. Lim**, J. Cho, and H. Yoon, “Generation of Solenoidal Modes and Magnetic Fields in Turbulence Driven by Compressive Driving,” *ApJ*, vol. 893, no. 1, p. 75, Apr. 2020, Art. no. 75.
 DOI: 10.3847/1538-4357/ab8066

Second- or third-author articles

- 1 Z.-Y. D. Lin, **J. Lim**, J. B. Simon, Z.-Y. Li, D. Carrera, M. Fernández-López, R. Harrison, R. Li, L. W. Looney, I. W. Stephens, and H. Yang, “Azimuthal Dust Polarization from Aerodynamically Aligned Grains as Evidence for the Streaming Instability in Protoplanetary Disks,” *ApJ*, vol. 1006, no. 1, p. 95, Jul. 2026, Art. no. 95.  DOI: 10.3847/1538-4357/ae8445
- 2 D. Carrera, L. E. J. Eriksson, **J. Lim**, W. Lyra, and J. B. Simon, “Positive feedback: II. How dust coagulation inside vortices can form planetesimals at low metallicity,” *A&A*, vol. 701, p. L1, Sep. 2025, Art. no. L1, **Contribution:** Participated in scientific discussions.  DOI: 10.1051/0004-6361/202555045
- 3 D. Carrera, **J. Lim**, L. E. J. Eriksson, W. Lyra, and J. B. Simon, “Positive feedback: How a synergy between the streaming instability and dust coagulation forms planetesimals,” *A&A*, vol. 696, p. L23, Apr. 2025, Art. no. L23, **Contribution:** Analyzed data from my streaming-instability simulations and participated in scientific discussions.  DOI: 10.1051/0004-6361/202554100

Other co-authored articles

- 1 S. A. Baronett, W. Lyra, H. Aly, O. Brouillette, D. Carrera, V. I. De Cun, L. E. J. Eriksson, M. Flock, P. Huang, L. Krapp, G. Lesur, R. Li, S. Li, **J. Lim**, S.-J. Paardekooper, D. G. Rea, D. Sengupta, J. B. Simon, P. Sudarshan, O. M. Umurhan, C.-C. Yang, and A. N. Youdin, “A Comparative Study of the Streaming Instability: Unstratified Models with Marginally Coupled Grains,” arXiv e-prints, arXiv:2603.04558, Mar. 2026, Art. no. arXiv:2603.04558, Performed simulations and participated in scientific discussions. arXiv: 2603.04558 [astro-ph.EP].
- 2 D. Godines, W. Lyra, L. Ricci, C.-C. Yang, J. B. Simon, **J. Lim**, and D. Carrera, “On the Mass Budget Problem of Protoplanetary Disks: Streaming Instability and Optically Thick Emission,” ApJ, vol. 997, no. 2, p. 192, Feb. 2026, Art. no. 192, **Contribution:** Participated in scientific discussions.  DOI: 10.3847/1538-4357/ae1f15

Seminar & Conference Presentations

- 1 **Dissertation talk**, The 247th American Astronomical Society Meeting, Phoenix, AZ, Jan. 2026.
- 2 **Contributed talk**, Star and Planet Formation Group, University of Nevada, Las Vegas, NV, Sep. 2025.
- 3 **Contributed talk**, TCAN Disk Hydro & Planet Formation, University of Arizona, Tucson, AZ, Nov. 2023.
- 4 **Contributed talk**, Magnetic Fields in the Universe 7, Quy Nhon, Vietnam, Feb. 2020.
- 5 **Contributed talk**, Korean Astronomical Society Meeting, Seoul, South Korea, Oct. 2019.